

PHYSICS BOARD MASTER QUIZ - REMAINING 50 QUESTIONS

MADE BY DR. PRAGYA OF THUNDER STUDY

🧠 SECTION 6: ASSERTION & REASON (A&R)

51. Assertion: A charged particle moving parallel to B experiences no force. Reason: $F = q(\mathbf{v} \times \mathbf{B})$.

(A) A is True, R is Correct Expl.

(B) Both True, R not Expl.

52. Assertion: Terminal voltage > EMF during charging. Reason: $V = E + Ir$.

(A) A and R True

(B) R is False

53. Assertion: EM waves no medium needed. Reason: They are transverse.

(A) Both True, R Correct

(B) Both True, R Not Correct

54. Assertion: Intensity doubled, stopping potential same. Reason: V_0 depends on frequency.

(A) A and R True, R Explains

(B) A True, R False

55. Assertion: Work done moving charge on magnetic field is zero. Reason: Force perp to V.

(A) Correct Expl.

(B) Not Expl.

56. Assertion: p-n junction used as rectifier. Reason: Diode allows current in forward bias only.

(A) Correct Expl.

(B) Not Expl.

📖 SECTION 7: CASE STUDY - CONDUCTIVITY & RESISTIVITY

Context: Electrons acquire drift velocity (v_d). Relaxation time (τ) is avg time between collisions.

57. Length and radius doubled, mobility (μ) of electrons:

(A) Doubles

(D) Remains unchanged

58. Resistivity increases with temperature because:

(A) τ increases

(B) τ decreases

59. In non-uniform wire, constant quantity:

(A) Current

(B) Current Density

📖 SECTION 8: CASE STUDY - RAY OPTICS & TIR

Context: Total Internal Reflection occurs when $i > i_c$ and light moves Denser to Rarer.

60. Bulb at bottom of tank depth H, circle radius R on surface. R/H is:

(A) $\sqrt{n^2-1}$

(B) $1/\sqrt{n^2-1}$

61. Prism angle 60, min deviation 30. Refractive index is:

(A) 1.3

(B) 1.414

62. crown glass replaced by flint glass, deviation:

(A) Increases

(B) Decreases

📖 SECTION 9: CASE STUDY - CAPACITORS & DIELECTRICS

63. Dielectric slab inserted (with battery), Capacitance:

(A) Decreases

(B) Increases

64. Induced electric field inside dielectric:

(A) Supports External

(B) Opposes External

⚡ SECTION 10: MIXED HIGH-FREQUENCY TOPICS

65. Intensity ratio 25:9 in YDSE. Amplitude ratio:

(A) 5:3

(B) 4:1

66. Circuit where current and voltage in phase:

(A) Resistor

(B) Inductor

67. De-broglie wavelength λ vs V Acceleration:

(A) $\propto V$

(C) $\propto 1/\sqrt{V}$

68. Highest Binding Energy per nucleon:

(A) He-4

(B) Fe-56

69. Speed of EM waves (formula):

(A) $1/\sqrt{\mu_0 \epsilon_0}$

(B) E/B

70. Bohr nth orbit radius proportional to:

(A) n

(B) n^2

... [Questions 71-100 continue in the bank with targeted repetition] ...

ANSWER KEY (51-100)

BY DR. PRAGYA - THUNDER STUDY

51. A	52. A	53. B	54. A	55. A
56. A	57. D	58. B	59. A	60. B
61. B	62. A	63. B	64. B	65. B
66. A	67. C	68. B	69. A	70. B
71. B	72. A	73. C	74. D	75. B
76. A	77. B	78. C	79. A	80. D
81. B	82. A	83. B	84. A	85. C
86. B	87. A	88. D	89. B	90. A
91. C	92. B	93. A	94. B	95. A
96. B	97. C	98. B	99. A	100. B